Supplementary Document

How Much of the Treasury Term Premium Is the Treasury?

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Not for publication

This document accompanies the replication package of the paper and its Internet Appendix. It holds the material that the main text does not rely on: the full tables behind the robustness checks that Internet Appendix Table IA.52 reports in one line, the panels of the appendix's tables that the main text does not use, the event-level detail of the policy announcements, and the documentation of the data. Section S.1 extends Part A of the Internet Appendix, Sections S.2 to S.5 extend Sections B.1 to B.4 in their order, and Section S.6 documents the data. Unless a caption says otherwise, the standard errors, intervals and sets are those of Internet Appendix Section A.9.

S.1 The Data and the Construction

This section extends Part A of the Internet Appendix, Sections A.2 to A.9, with the checks and the detail by measure that the appendix summarizes.

S.1.1 The OIS Record

This subsection extends Internet Appendix Section A.2. The series of record runs from February 2002 to June 2026, 293 months at 2 and 5 years, and from July 2008, when the 10-year quote begins, 198 months at 10. Set beside it on the same months, the spread on Bloomberg's SOFR OIS rate has era means that differ by at most 1.9 basis points over May 2019 to December 2021 and by at most 3.3 since 2022. In that era the SOFR OIS rate lies above the federal funds OIS rate on average at every maturity, and in single months the two differ by at most 6.9.

The month-end synthetic OIS rate built from traded prices exists in 89 of the 96 months of 2013 to 2020 at 5 years and in 88 at 10 (Internet Appendix Section A.2.2). Its mean difference from the quote at 10 years over 2013 to 2017, 2.6 basis points, comes from a few days. The synthetic stands more than 20 basis points from the quote on 19 of 1,190 days. On those days the hybrid, built from the same day's basis trades and the close of the LIBOR swap, stays within 1.6 basis points of the quote. With the synthetic in place of the quote at 10 years, the OIS swap spread and $s - d$ move by 0.3 to 0.8 basis points in 2013. Over 2014

to 2020 they move by 3.1 on EIB and KfW, 3.4 on Farm Credit and 0.5 on the Home Loan Banks (Panel B of Internet Appendix Table IA.5).

Before 2008 I set the quoted OIS rate against a synthetic rate (Internet Appendix Section A.2.3). It is the LIBOR swap rate, put on the OIS's annual actual/360 fixed leg, less the trailing 12-month mean of the spread of 3-month LIBOR over the federal funds rate. At the 77 month-ends of February 2002 to June 2008 the synthetic rate lies 3.6 basis points below the quoted 5-year OIS rate on average and 3.1 below the 2-year, with a standard deviation of 13 to 14. Their monthly changes correlate at 0.99. With the trailing 3-month mean of the spread the 5-year gap is 6.2. Starting the first era in January 2005 instead of February 2002 raises its mean OIS swap spread from 32.3 to 38.9 basis points. The narrowing of the advantage over the Home Loan Banks is 32.4 basis points on the first start and 32.8 on the second (Internet Appendix Table IA.6).

S.1.2 The Comparators

This subsection extends Internet Appendix Section A.4. How much of the Treasury's yield advantage over EIB and KfW is the cross-currency basis? In Table S.1 I give the slopes of the advantage and of $z = (s - y) - \hat{\chi}$ on the basis, with their fixed-b critical values. I also give the era means from which I form, in Internet Appendix Table IA.9, the change in z with the basis held at its mean. At 5 years the basis was negative in 182 of the 183 months from April 2011 to June 2026 and averaged -22.3 basis points. At 10 years the slope of the advantage on the basis is 0.37, with a standard error of 0.17, and it is not distinguishable from zero at the 5% level against the fixed-b critical value.

For Farm Credit I have 447 non-callable fixed-rate bullets near 5 years and 463 near 10 from February 2002. On the Home Loan Banks' bullets one value of the solved yield less the coupon, -306.9 basis points, comes from a reopening of an older bond with a high coupon sold well above par. I set it aside in Internet Appendix Section A.4.2. With it the solved yield less the coupon averages 0.15 basis points at 5 years, and without it 0.59.

Regressed on the Treasury yield of the same maturity from April 2011, the yield of EIB and KfW less the Home Loan Banks' falls by 3.0 basis points per percentage point at 5 years and by 10.6 at 10. Less Farm Credit's it falls by 0.8 and 4.6. Each slope has a fixed-b interval that contains zero. At the upper ends of those intervals the tax wedge since 2022 is at most 6.5 basis points at 5 years against the Home Loan Banks and 11.1 against Farm Credit. At 10 years the agencies' monthly series correlate with EIB and KfW's at 0.46 to 0.78 on the months they share since 2014, against -0.11 to 0.52 at 5 years. Before 2008 the Treasury's 5-year advantage over the agencies, 40.2 and 37.2 basis points, is 2.3 and 2.1

Table S.1. EIB and KfW and the cross-currency basis: slopes and era means

In basis points, month-end, April 2011 to June 2026. $\hat{\chi}$, the Treasury's yield advantage, is EIB and KfW's dollar yield less the constant-maturity Treasury yield, $z = (s - y) - \hat{\chi}$, and x the quoted EUR/USD basis on the euro leg. Panel A regresses z on x (A), with era intercepts (B), and $\hat{\chi}$ on x (C), with the fixed-b critical value (Kiefer and Vogelsang, 2005); in Panel B, (A) and (B) are z at the mean basis on each slope. Internet Appendix Table IA.9 gives the change in z into each era. Inference as in Internet Appendix Section A.9.

Panel A: Slopes on the basis

Maturity	Months	Lag	(A)	(B)	(C)	5% critical value
5 years	183	13	0.68 (0.22)	0.67 (0.20)	−0.78 (0.22)	2.19
10 years	165	13	0.45 (0.29)	0.57 (0.29)	−0.37 (0.17)	2.21

Panel B: Era means at 5 years

Era	Months	Basis x	$\hat{\chi}$	z	z, (A)	z, (B)
Apr 2011 to Dec 2013	33	−31.4	37.0	−47.7	−41.5	−41.6
Jan 2014 to Dec 2021	96	−23.2	13.0	−32.5	−31.9	−31.9
Jan 2022 to Jun 2026	54	−15.0	4.5	−34.1	−39.1	−39.1

times the convenience of the 3-month bill over general collateral repo, 17.7 basis points.

The European Investment Bank's 3-year note of 19 February 2013 is one example of the documentation in Internet Appendix Section A.4.4. It opened with guidance of mid-swaps plus 7 basis points and was priced at 23.9 basis points over the Treasury note due 15 February 2016.

Does any further measure of the Treasury's yield advantage change its level or its fall? In Table S.2 I set seven measures side by side over May 2011 to February 2024, with the drivers of five of them. They are the three comparators, the curves of the European Investment Bank and of KfW alone, Bloomberg's curve of Fannie Mae and the strip curve of the Resolution Funding Corporation. I recover the Fannie Mae curve as the strip curve's spread over the Treasury less its spread over the Fannie Mae curve, both par against par, so the Fannie Mae measure shares its pricing source with the strips'. I give the reading in Internet Appendix Section A.4.3. At 5 years every measure but Refcorp's lies between 6.9 and 14.2 basis points over 2014 to 2021 and between 3.0 and 11.2 over January 2022 to February 2024. From the first window to the last the Treasury's 5-year advantage fell by 13.1 to 40.7 basis points on EIB and KfW, on each curve alone and on the two agencies. It rose on the two measures read from the strip curve, Fannie Mae's from 8.5 to 11.2 and Refcorp's from 46.8 to 51.2. Both contain the strip curve's spread over the Treasury, so the two rises are not independent of each other. None of the five curves follows the convenience of the 3-month bill closely, the correlations at 5 years running from 0.06 to 0.15 in levels.

Table S.2. Seven measures of the Treasury's yield advantage from three families, May 2011 to February 2024

In basis points. Each measure is a dollar yield less the constant-maturity Treasury yield of the same maturity at month-end: EIB and KfW together, the first comparator of Internet Appendix Table IA.8, and each curve alone; Bloomberg's curve of Fannie Mae, recovered as the Refcorp strip curve's spread over the Treasury less its spread over the Fannie Mae curve, both par against par; the Resolution Funding Corporation (Refcorp) strip curve on the par convention; and the two dollar comparators on their own months, which the columns of months count. Panel A gives the window means and the excess of the Treasury yield over the benchmark yield, $\theta - \hat{\chi}$, on the range of θ I use in the era that contains the window (main-text Table 6 at 5 years, Internet Appendix Table IA.32 at 10). Panel B gives each curve's loadings on the log of marketable debt over GDP and on the log of the VIX in a regression in levels that also holds the bill share of marketable debt, with Newey and West (1987) standard errors at 12 lags in parentheses, and the correlation of its 5-year measure with the convenience of the 3-month bill, general collateral repo less the bill on the money-market basis.

Panel A: Window means and the Treasury yield less the benchmark yield (bp)

	Months		$\hat{\chi}$		$\theta - \hat{\chi}$	
Measure	5 years	10 years	5 years	10 years	5 years	10 years
May 2011 to December 2013						
EIB and KfW	32	32	37.7	51.5	[−29.1, −0.5]	[−38.3, 23.0]
EIB	32	32	46.5	63.3	[−37.9, −9.2]	[−50.2, 11.2]
KfW	32	32	29.0	39.6	[−20.4, 8.3]	[−26.5, 34.8]
Fannie Mae	32	32	8.5	24.7	[0.1, 28.8]	[−11.5, 49.8]
Refcorp	32	32	46.8	48.1	[−38.2, −9.5]	[−34.9, 26.4]
Farm Credit	27	32	20.5	40.5	[−11.9, 16.8]	[−27.4, 33.9]
Home Loan Banks	27	22	19.5	39.7	[−10.9, 17.7]	[−26.6, 34.8]
January 2014 to December 2021						
EIB and KfW	96	96	13.0	15.5	[5.0, 21.8]	[12.3, 54.2]
EIB	96	96	14.2	15.7	[3.9, 20.7]	[12.0, 54.0]
KfW	96	96	11.9	15.2	[6.2, 22.9]	[12.5, 54.5]
Fannie Mae	96	96	7.9	30.5	[10.1, 26.9]	[−2.8, 39.2]
Refcorp	96	96	45.1	54.8	[−27.1, −10.3]	[−27.1, 14.9]
Farm Credit	86	89	9.4	35.4	[8.7, 25.4]	[−7.7, 34.3]
Home Loan Banks	77	43	6.9	31.5	[11.1, 27.9]	[−3.8, 38.2]
January 2022 to February 2024						
EIB and KfW	26	26	7.3	11.8	[23.8, 27.0]	[36.0, 56.9]
EIB	26	26	5.8	9.3	[25.4, 28.5]	[38.5, 59.4]
KfW	26	26	8.9	14.3	[22.3, 25.5]	[33.5, 54.4]
Fannie Mae	26	26	11.2	30.2	[20.0, 23.1]	[17.6, 38.5]
Refcorp	26	26	51.2	68.4	[−20.0, −16.8]	[−20.6, 0.3]
Farm Credit	26	19	3.0	44.4	[28.1, 31.3]	[3.4, 24.3]
Home Loan Banks	24	17	6.4	45.0	[24.8, 28.0]	[2.7, 23.6]

Panel B: The drivers of the advantage and the bill, the five curves over the 154 months

	5 years, levels		10 years, levels		Correlation with the bill	
Measure	Log debt/GDP	Log VIX	Log debt/GDP	Log VIX	Levels	12-month changes
EIB and KfW	−119.6 (34.8)	17.7 (5.1)	−157.7 (45.1)	19.9 (5.2)	0.12	0.20
EIB	−154.7 (48.0)	21.8 (7.0)	−204.3 (60.8)	26.0 (7.3)	0.10	0.17
KfW	−84.5 (21.8)	13.6 (3.5)	−111.2 (30.2)	13.7 (3.7)	0.15	0.23
Fannie Mae	−9.6 (8.4)	13.3 (2.2)	21.3 (21.4)	15.5 (3.7)	0.06	0.08
Refcorp	−21.5 (22.3)	13.7 (5.1)	36.1 (26.7)	12.5 (6.6)	0.12	0.22

S.1.3 The Survey Term Premia

This subsection extends Internet Appendix Section A.5.1. Do the survey term premia depend on how I take the surveys? In Table S.3 I give three checks. The first averages the rates over the last seven days of the month, the second uses the Survey of Professional Forecasters alone, and the third sets the Blue Chip long-range survey against the extension that I use beyond the sixth quarter. In the 13 issues that contain the long-range survey, the long-range construction lies above the extension by 6.5 basis points on average for the 5-year OIS-curve premium and 4.0 for the Treasury-curve premium. In a single issue it lies above by up to 50. A fourth check, which the table does not give, takes the long-range survey's spread of the funds rate over the bill for the calendar year, from the latest issue before the survey month, in place of the Blue Chip spread of the sixth quarter. I give its result since 2022 in Internet Appendix Section A.5. On the 37 months from November 2018 to December 2021 that the check covers, the Treasury curve's 5-year premium exceeds the OIS curve's by 17.5 basis points, against 16.2.

S.1.4 The Price of the Sovereign's Credit

This subsection extends Internet Appendix Section A.6. The bill market's measure of the credit premium is e_t of Internet Appendix Section A.6.2. In the debt-limit episode of 2023 it averaged 21.3 basis points and exceeded 10 basis points on 57 days, against 169 days outside the episodes since 2022.

S.1.5 The Funding Commitment Price

This subsection extends Internet Appendix Sections A.7.2 to A.7.4 with the detail of the samples behind the funding commitment price. Without the notes of more than 4.5 years, 2,967 notes remain and only one lies above 4 years. In Panel C of Internet Appendix Table IA.16 I therefore compare the two shapes that can still be estimated with the same shapes on all notes. At 5 years both ends of the range rise by 1.3 to 1.8 basis points in every era. At 2 years the ends move by less than half a basis point, and at 10 years they rise by 3.6 to 6.4. The two cooperatives' gradient is 6.99 basis points per year on their own panel and 6.38 on the five-issuer panel. The gap is composition, since on the 2,156 notes that both panels hold the two record the same spread.

To put the Home Loan Banks' record on one index, the federal funds rate, I adjust the spread at issue of each note indexed to 3-month LIBOR. I add to it the term spread of the 3-month LIBOR swap over the federal funds OIS at the note's maturity on its issue date,

Table S.3. Three checks of the survey term premia

In basis points. The first rows give the era means of the two survey premia of Internet Appendix Table IA.12 and of their difference, the Treasury curve's premium less the OIS curve's, with each rate taken as its average over the last seven days of the month before the issue instead of at the month-end; the eras are those of Internet Appendix Table IA.12. The next row gives the Treasury-curve premium at 5 years on the Survey of Professional Forecasters alone, on its own survey months, whose number is in brackets: its quarterly bill path for the current quarter and the next five, then its annual forecasts and its 10-year average, with the Treasury yield at the end of the first month of the survey's quarter. The last rows give, in the 13 issues that contain the Blue Chip long-range survey, the long-range construction less the extension, its mean and its largest absolute value.

Check	Maturity	Feb 2002 to Jun 2008	Jul 2008 to Dec 2013	Jan 2014 to Dec 2021	Jan 2022 to Jan 2025
Last seven days: Treasury curve	2 years	−15.5	−10.9	−14.0	19.5
Last seven days: OIS curve	2 years	−4.1	−14.9	−24.9	3.2
Last seven days: difference	2 years	−11.4	4.0	10.9	16.3
Last seven days: Treasury curve	5 years	13.3	−30.2	−28.4	56.6
Last seven days: OIS curve	5 years	33.2	−31.6	−51.7	21.1
Last seven days: difference	5 years	−20.0	1.4	23.3	35.5
Survey of Professional Forecasters alone: Treasury curve	5 years	19.8 [25]	−20.7 [22]	−24.0 [32]	48.2 [18]

		Treasury curve		OIS curve	
		Mean	Largest absolute	Mean	Largest absolute
Long-range survey less the extension, 13 issues	2 years	2.3	9.5	3.1	9.3
	5 years	4.0	49.0	6.5	49.8

interpolated between the quoted maturities. That spread averages 29.8, 45.7 and 30.0 basis points over the notes of the three eras before 2022. Notes indexed to the funds rate or to SOFR enter as they are. Notes on other indices, most of them on 1-month LIBOR, leave the sample, as do the 12 on 3-month LIBOR with no quoted term spread at their maturity. The record then holds 199 identifying notes before 2008, 429 over 2008 to 2013 and 1,384 over 2014 to 2021. Without the notes indexed to SOFR the gradient on one index over 2014 to 2021 is 9.82 basis points per year. Where the one-index gradient lies above the other two measures, the range I use extends to the upper end of the one-index measure, 46.6 basis points at 5 years over 2008 to 2013 and 37.7 over 2014 to 2021. On the higher of the other two measures the upper ends are 37.2 and 34.8 at 5 years, and 18.9 and 15.3 at 2 years.

Before 2008 the 5-year set of $s - d$ on each era's own notes lies above zero on both agencies, $[1.0, 48.8]$ on Farm Credit and $[3.7, 50.4]$ on the Home Loan Banks. With the index held at SOFR it contains zero, $[-26.6, 39.3]$ and $[-24.4, 40.4]$, and so does the set across the two measures that I report in Internet Appendix Section A.8.

S.1.6 The Benchmark Yield by Comparator and Measure

This subsection extends Internet Appendix Section A.8 with the era means behind Internet Appendix Table IA.20. In Tables S.4 and S.5 I give the accounting by era on the Home Loan Banks and on every comparator and every measure of the funding commitment price. From the era before 2008 to the era since 2022 the OIS swap spread fell by 59.4 basis points on the Home Loan Banks' months and 61.3 on Farm Credit's, and the Treasury's advantage fell by 32.4 and 40.7. From 2014 to 2021 to the era since 2022 the advantage fell by 2.1 basis points on the Home Loan Banks, 9.9 on Farm Credit and 8.5 on EIB and KfW, and the sets of the last two exclude zero.

S.1.7 The Length of the Block

This subsection extends Internet Appendix Section A.9. Do the bootstrap sets depend on the length of the block? In Table S.6 I give the sets with blocks of 12, 24 and 36 months, and the robustness summary of Internet Appendix Table IA.52 gives the comparison in one row. Since 2022 the upper end of the fixed-b interval of the advantage lies 9 to 16 basis points below the lower end of the funding commitment price on every comparator. The excess of the Treasury yield over the benchmark yield since 2022 is therefore positive under either inference. From the era before 2008 to the era since 2022 the fall in the advantage is 55% of the fall in the spread on the Home Loan Banks, with a 95% set of 42% to 69%. On Farm Credit it is 66%, with a set of 49% to 85%.

Table S.4. The Treasury yield against the benchmark yield at 5 years on the Home Loan Banks

In basis points. $s - y$ is the OIS swap spread; $\hat{\chi} = y^m - y$ is the Treasury's yield advantage over the Home Loan Banks' fixed-rate bullets; and θ is the funding commitment price of main-text Table 6, the range used across its measures. The benchmark is $d = y^m - \theta$. The excess of the Treasury yield over the benchmark yield is $y - d = \theta - \hat{\chi}$, and the dealers' charge net of the comparator's premium is $s - d = (s - y) + \theta - \hat{\chi}$, so that the two differ by $y - s$ (Internet Appendix Section A.8). Each era mean is taken over the months on which $s - y$ and $\hat{\chi}$ both exist. Each cell gives the range across the measures of θ and, below it in brackets, the 95% confidence set from a circular moving-block bootstrap (12-month blocks, 4,999 replicates) in which θ is drawn on every replicate with the other inputs, from the sampling distributions of the maturity gradient, the share of it that is the agencies' own credit and the ratio of each era's gradient; the set is the union of the sets across the measures of θ, from the 2.5th percentile at the lowest measure to the 97.5th at the highest. The last rows give the change from the era before 2008 to the era since 2022 for each measure of θ before 2022, on each era's own notes and with the reset index held at SOFR, with the 95% set of a circular moving-block bootstrap of the two eras. I give the accounting on Farm Credit and on EIB and KfW in Internet Appendix Table S.5.

Era	Months	$s - y$	$\hat{\chi}$	θ	$y - d$	$s - d$
Feb 2002 to Jun 2008	48	31.4	37.2	11.9 to 33.5	−25.3 to −3.7	6.0 to 27.7
					[−56.1, 18.6]	[−24.4, 50.4]
Jul 2008 to Dec 2013	55	−1.2	37.0	8.6 to 46.6	−28.5 to 9.6	−29.6 to 8.4
					[−57.1, 35.8]	[−50.8, 29.7]
Jan 2014 to Dec 2021	77	−19.2	6.9	18.1 to 37.7	11.1 to 30.8	−8.1 to 11.6
					[6.3, 40.2]	[−15.4, 22.5]
Jan 2022 to Jun 2026	39	−28.0	4.9	31.2 to 34.3	26.3 to 29.5	−1.7 to 1.5
					[20.9, 34.9]	[−7.6, 7.2]
Change from February 2002 to June 2008 to January 2022 to June 2026						
Own notes	48, 39	−59.4	−32.4	0.8 to 0.8	33.1 to 33.2	−26.2 to −26.2
					[11.6, 56.5]	[−48.1, −2.9]
Index at SOFR	48, 39	−59.4	−32.4	19.3 to 21.2	51.6 to 53.6	−7.7 to −5.8
					[20.1, 87.8]	[−38.7, 27.7]

Table S.5. The Treasury yield against the benchmark yield at 5 years on every comparator

In basis points. $s-y$ is the OIS swap spread; $\hat{\chi} = y^m - y$ is the Treasury's yield advantage over the comparator (Internet Appendix Table IA.8); and θ is the funding commitment price of main-text Table 6, the range used across its measures. The benchmark is $d = y^m - \theta$, the excess of the Treasury yield over the benchmark yield is $y - d = \theta - \hat{\chi}$, and the dealers' charge net of the comparator's premium is $s - d = (s - y) + \theta - \hat{\chi}$. Each era mean is taken over the months on which $s - y$ and $\hat{\chi}$ both exist. Each cell gives the range across the measures of θ and, below it in brackets, the 95% confidence set from a circular moving-block bootstrap (12-month blocks, 4,999 replicates) in which θ is drawn on every replicate, the union of the sets across the measures of θ. The rows of the Home Loan Banks are those of Table S.4. Over July 2008 to December 2013 EIB and KfW enter from April 2011.

Comparator	Months	$s - y$	$\hat{\chi}$	θ	$y - d$	$s - d$
February 2002 to June 2008						
Farm Credit	58	31.8	40.2	11.9 to 33.5	−28.3 to −6.7	3.5 to 25.1
					[−59.7, 17.5]	[−26.6, 48.8]
Home Loan Banks	48	31.4	37.2	11.9 to 33.5	−25.3 to −3.7	6.0 to 27.7
					[−56.1, 18.6]	[−24.4, 50.4]
July 2008 to December 2013						
EIB and KfW	33	−10.7	37.0	8.6 to 46.6	−28.4 to 9.6	−39.1 to −1.1
					[−49.8, 31.0]	[−59.9, 19.6]
Farm Credit	48	−2.9	32.8	8.6 to 46.6	−24.2 to 13.9	−27.1 to 10.9
					[−48.0, 36.1]	[−45.4, 29.5]
Home Loan Banks	55	−1.2	37.0	8.6 to 46.6	−28.5 to 9.6	−29.6 to 8.4
					[−57.1, 35.8]	[−50.8, 29.7]
January 2014 to December 2021						
EIB and KfW	96	−19.5	13.0	18.1 to 37.7	5.0 to 24.7	−14.5 to 5.2
					[−1.0, 34.9]	[−24.2, 17.9]
Farm Credit	86	−19.4	9.4	18.1 to 37.7	8.7 to 28.3	−10.8 to 8.9
					[3.4, 38.2]	[−18.6, 20.5]
Home Loan Banks	77	−19.2	6.9	18.1 to 37.7	11.1 to 30.8	−8.1 to 11.6
					[6.3, 40.2]	[−15.4, 22.5]
January 2022 to June 2026						
EIB and KfW	54	−29.6	4.5	31.2 to 34.3	26.7 to 29.8	−3.0 to 0.2
					[23.6, 33.1]	[−7.2, 4.7]
Farm Credit	51	−29.5	−0.5	31.2 to 34.3	31.6 to 34.8	2.2 to 5.4
					[25.9, 39.7]	[−3.2, 11.6]
Home Loan Banks	39	−28.0	4.9	31.2 to 34.3	26.3 to 29.5	−1.7 to 1.5
					[20.9, 34.9]	[−7.6, 7.2]

Table S.6. The bootstrap sets of the era means by the length of the block

In basis points, at 5 years, on each comparator's matched months: the 95% sets of the circular moving-block bootstrap of Internet Appendix Table IA.20 with blocks of 12, 24 and 36 months, for the excess of the Treasury yield over the benchmark yield, $y - d$, in each era since 2014, the union across the measures of θ, and for the changes from the era before 2008 to the era since 2022 with θ on each era's own notes.

The 95% set by the length of the block					
		Months	12 months	24 months	36 months
$y - d$, January 2014 to December 2021					
Home Loan Banks		77	[6.3, 34.6]	[6.3, 34.3]	[6.5, 34.4]
Farm Credit		86	[3.4, 32.6]	[3.7, 32.1]	[4.1, 31.8]
EIB and KfW		96	[−1.0, 29.5]	[−1.9, 29.8]	[−2.0, 30.1]
$y - d$, January 2022 to June 2026					
Home Loan Banks		39	[20.9, 34.9]	[22.0, 33.9]	[23.5, 32.4]
Farm Credit		51	[25.9, 39.7]	[26.8, 39.6]	[27.5, 39.2]
EIB and KfW		54	[23.6, 33.1]	[23.4, 33.2]	[23.8, 33.0]
Change from February 2002 to June 2008 to January 2022 to June 2026, θ on each era's own notes					
Home Loan Banks	$\hat{\chi}$	48, 39	[−42.3, −24.4]	[−40.0, −24.6]	[−38.5, −26.2]
	$y - d$		[11.6, 56.5]	[12.3, 56.1]	[13.3, 54.9]
	$s - d$		[−48.1, −2.9]	[−47.0, −3.7]	[−45.9, −3.8]
Farm Credit	$\hat{\chi}$	58, 51	[−53.5, −28.5]	[−52.7, −30.4]	[−50.7, −30.6]
	$y - d$		[18.5, 65.7]	[19.9, 65.9]	[19.9, 64.6]
	$s - d$		[−41.6, 3.7]	[−42.0, 4.3]	[−41.1, 3.3]

S.2 The Difference between the Term Premia on the Two Curves

This section extends Internet Appendix Section B.1 with the survey test at the lag of Schwert (1989) and the return regression on the lagged spread.

In the other currencies of Internet Appendix Table IA.22, Bloomberg's euro curve before October 2019 is the EONIA curve less 8.5 basis points. Over 2014 to 2021 and since 2022 the 10-year OIS rate less the government yield is negative in sterling, at −19.6 and −24.5 basis points, as it is in dollars, at −28.5 and −40.5. It is close to zero in yen, at 1.3 and 2.0, and positive in the euro, at 10.1 and 11.9, and in the Swiss franc, whose sample begins in August 2017, at 18.2 and 44.8. Each of these means is distinguishable from zero at the 5% level except the yen's in both eras and the euro's since 2022.

The gap between a published premium and the same premium read on the OIS curve is the same for both models (Internet Appendix Section B.1.2). The OIS swap spread accounts for 84% to 93% of its variance in levels in every era from 2008, and for 42% to 53% in

monthly changes. The survey premium differs from each published model by more than the two models differ from each other, with era means of the difference from −124 to 102 basis points. Of the principal components of the zero curve of Gürkaynak et al. (2007) at 3 to 120 months, the first accounts for 98.3% of their variance and the first five for more than 99.99% (Internet Appendix Section B.1.3). When I extract the components on each regression's own sample, no R^2 with five components moves by more than 0.005.

Does the OIS swap spread move with revisions of the expected spread of the funds rate over the bill? In Table S.7 I give the slope of the monthly change in the spread on the revision between consecutive Blue Chip issues, at month-ends and on the survey window (Internet Appendix Section B.1.5). Beside each slope are the loadings that a spread holding the expectation would show without persistence and with full persistence. In levels the 2-year spread moves with the expected funds-bill spread, and the 10-year spread moves against it (Panel C). The level of the 2-year spread loads 84.4 basis points per percentage point on the level of the expected funds-bill spread at month-end and 91.2 on the survey window. Both intervals exclude zero and contain the 50 that holding the expectation without persistence implies. At 10 years it loads −96.1 and −102.7, with intervals that also exclude zero, and at 5 years both intervals contain zero. These level loadings come mostly from the differences between eras. In Table S.8 I regress the Treasury's return relative to the swap on the spread lagged one month (Internet Appendix Section B.1.6). A component of the month-end spread that reverts the next month does not move that regressor.

S.3 The Treasury's Yield Advantage and the Agency Gap

This section extends Internet Appendix Section B.2 with the break tests not used in the main text, the gap on the realized path of the short end and the 10-year accounting on the synthetic OIS rate. In Table S.9 I give the breaks in the mean of the Treasury's advantage that the procedure of Bai and Perron (2003) finds, and the mean of each segment.

The steps at each date and the changes across each span are in Panels A and B of Internet Appendix Table IA.28 (Internet Appendix Section B.2.1). At the onset of the euro-area crisis in July 2011 the advantage rose by 4.9 basis points on the Home Loan Banks and 10.1 on Farm Credit. After August 2012 it fell by 10.5 basis points on the Home Loan Banks, 11.5 on Farm Credit and 29.9 on EIB and KfW, each distinguishable from zero at the 1% level. At the final rule of the liquidity coverage ratio in September 2014 no step is distinguishable from zero at the 5% level, and at the rule's phase-in to 80% in January 2015 only Farm Credit's, −4.3 basis points, is. At the reform of money-market funds in October 2016 the agencies' advantage fell by 2.8 and 5.3 basis points. At the rule's full phase-in in January 2017 the

Table S.7. The survey test at the Schwert lag

In basis points per percentage point. Panel A gives the slope of the monthly change in the OIS swap spread on the revision between consecutive Blue Chip issues of the expected funds-bill spread, over the next four quarters or in the farthest quarter the two issues share, from March 2002 to January 2025, at the month-end before each issue or on the average of the last seven days of that month, which I call the survey window. Newey and West standard errors at the Schwert lag are in parentheses, beside the fixed-b 95% interval (Bartlett kernel, no prewhitening). If the N-year spread held the expected average of the funds-bill spread over its life, a revision of one percentage point over the next four quarters would move it by $100/N$ basis points if the revision did not persist beyond them and by $100(4N-1)/(4N)$ if it persisted to maturity, and a revision of the farthest quarter alone by $25/N$ and $100(N-1)/N$; the last two columns give these loadings. Panel B gives the responses of the OIS rate, the Treasury yield and the spread to the revision of the expected funds-rate path over the next four quarters. Panel C gives the slope of the level of the spread on the level of the expected funds-bill spread over the next four quarters, at the Schwert lag, on 273 months at 2 and 5 years and 181 at month-end and 178 on the survey window at 10, and the slope of the change in the spread between the 13 long-range issues on the revision of their five-year averages of the expected funds-bill spread, at two lags, since the Schwert rule would set seven lags on 12 changes.

Panel A: The spread on the revision of the expected funds-bill spread

			Month-end		Survey window		Implied loading	
Revision	Maturity	Months	Slope	95% interval	Slope	95% interval	No persistence	Full persistence
Next four quarters	2 years	273	22.2 (18.6)	[−17.5, 61.9]	37.6 (19.0)	[−2.9, 78.2]	50.0	87.5
	5 years	273	−19.1 (9.5)	[−39.3, 1.1]	−3.2 (6.9)	[−18.0, 11.6]	20.0	95.0
	10 years	178	−3.3 (15.1)	[−36.4, 29.8]	18.4 (13.3)	[−10.9, 47.7]	10.0	97.5
Farthest quarter	2 years	273	18.7 (17.2)	[−18.1, 55.5]	30.2 (15.2)	[−2.1, 62.6]	12.5	50.0
	5 years	273	−11.3 (10.2)	[−33.2, 10.5]	−0.3 (7.6)	[−16.6, 16.0]	5.0	80.0
	10 years	178	−10.0 (12.4)	[−37.2, 17.2]	0.4 (9.6)	[−20.6, 21.5]	2.5	90.0

Panel B: The two rates and the spread on the revision of the expected funds rate

		Month-end			Survey window		
Maturity	Months	OIS rate	Treasury yield	Spread	OIS rate	Treasury yield	Spread
2 years	273	82.3 (7.6)	76.8 (8.0)	5.5 (2.2)	89.1 (5.9)	82.8 (6.7)	6.3 (2.5)
5 years	273	69.1 (8.1)	67.2 (8.3)	1.9 (1.3)	76.3 (7.4)	73.5 (7.7)	2.7 (1.4)
10 years	178	57.9 (9.6)	55.2 (10.2)	2.7 (2.4)	63.6 (8.3)	60.1 (9.6)	3.5 (2.3)

Panel C: Levels and the long-range issues

		Month-end		Survey window	
Regression	Maturity	Slope	95% interval	Slope	95% interval
Levels	2 years	84.4 (28.8)	[23.0, 145.8]	91.2 (27.9)	[31.5, 150.8]
	5 years	76.4 (59.1)	[−49.8, 202.6]	81.2 (58.9)	[−44.6, 206.9]
	10 years	−96.1 (40.2)	[−184.2, −8.0]	−102.7 (34.6)	[−178.7, −26.8]
Long-range issues, 12 changes	5 years	55.5 (58.8)	[−104.4, 215.3]	21.3 (48.5)	[−110.5, 153.1]
	10 years	26.1 (45.9)	[−98.8, 151.0]	−6.4 (33.4)	[−97.2, 84.4]

Table S.8. The Treasury's return relative to the swap on the spread lagged one month

The dependent variable is that of Internet Appendix Table IA.26. The table regresses it on $(y-s)_{t-1}$, the spread one month before the return starts, and, in the row marked instrumented, on $(y-s)_t$ instrumented by $(y-s)_{t-1}$; a component of the month-end spread that reverts the next month moves neither. Standard errors are Newey-West at $h-1$ lags with the fixed-b 95% interval of Kiefer and Vogelsang (2005), and at the larger of h and the Schwert (1989) lag with its fixed-b interval.

h	Sample	Design	Months	Slope (s.e.)	Interval	Interval, floor	Carry
3	all months	Lagged	289	0.32 (0.12)	[0.08, 0.56]	[0.03, 0.60]	0.25
	all months	Instrumented	289	0.32 (0.12)	[0.08, 0.57]	[0.03, 0.62]	0.25
	Feb 2002 to Jun 2008	Lagged	76	0.65 (0.44)	[−0.26, 1.56]	[−0.33, 1.63]	0.25
	Jul 2008 to Dec 2013	Lagged	66	1.13 (0.40)	[0.29, 1.97]	[0.61, 1.65]	0.25
	Jan 2014 to Dec 2021	Lagged	96	1.37 (0.45)	[0.45, 2.30]	[0.35, 2.39]	0.25
	Jan 2022 to Jun 2026	Lagged	51	1.21 (0.49)	[0.17, 2.24]	[−0.32, 2.73]	0.25
6	all months	Lagged	286	0.65 (0.25)	[0.14, 1.15]	[0.08, 1.21]	0.50
	all months	Instrumented	286	0.66 (0.26)	[0.14, 1.18]	[0.07, 1.24]	0.50
	Feb 2002 to Jun 2008	Lagged	76	1.07 (0.72)	[−0.51, 2.65]	[−0.49, 2.64]	0.50
	Jul 2008 to Dec 2013	Lagged	66	2.10 (0.39)	[1.23, 2.96]	[1.35, 2.84]	0.50
	Jan 2014 to Dec 2021	Lagged	96	2.78 (0.80)	[1.07, 4.50]	[1.16, 4.41]	0.50
	Jan 2022 to Jun 2026	Lagged	48	1.68 (0.50)	[0.51, 2.85]	[0.43, 2.93]	0.50
12	all months	Lagged	280	1.23 (0.43)	[0.34, 2.12]	[0.29, 2.16]	1.00
	all months	Instrumented	280	1.25 (0.44)	[0.33, 2.17]	[0.28, 2.22]	1.00
	Feb 2002 to Jun 2008	Lagged	76	3.08 (1.32)	[−0.14, 6.29]	[−0.14, 6.29]	1.00
	Jul 2008 to Dec 2013	Lagged	66	2.56 (0.38)	[1.60, 3.53]	[1.59, 3.53]	1.00
	Jan 2014 to Dec 2021	Lagged	96	5.16 (0.80)	[3.28, 7.03]	[3.23, 7.09]	1.00
	Jan 2022 to Jun 2026	Lagged	42	2.21 (0.82)	[−0.10, 4.52]	[−0.17, 4.59]	1.00

advantage fell on every comparator, by 6.5 to 7.2 basis points, each distinguishable from zero at the 5% level. At the conservatorship of Fannie Mae and Freddie Mac in September 2008 the agencies' steps, 18.5 and 0.7 basis points, are not distinguishable from zero at the 10% level.

From the twelve months before September 2014 to the twelve after January 2017, the advantage fell by 7.4 basis points on the Home Loan Banks, 12.6 on Farm Credit and 1.3 on EIB and KfW. These are 23%, 31% and 4% of the changes between the outer eras, −32.4 and −40.7 on the two agencies and −33.2 on EIB and KfW, whose first era runs from May 2011 to December 2013. Over spans of the same length elsewhere in the sample the advantage falls by as much or more 34%, 27% and 73% of the time. Across the euro-area crisis, from July 2011 to August 2012, the agencies' advantage fell by 7.5 and 4.9 basis points.

On the strip curves (Internet Appendix Section B.2.2), the advantage over EIB and KfW less the advantage over the strips fell by 34.8 basis points, with a standard error of 4.8. The bill measure's 95% interval of the rise of the sovereign's credit reaches 14.6 basis points, the higher of the two measures' upper ends. With the strips' own premium unchanged and a rise of the sovereign's credit that large, EIB and KfW's own premium still fell by 20.1 basis points. At 10 years, where the later window's point lies beyond the last strip and is the vendor's extrapolation, the advantage over the strips rose by 20.4 basis points and the advantage over EIB and KfW fell by 39.7. The advantage over Fannie Mae's curve, read from the same strips, rose by 2.7 basis points at 5 years.

The last of the Resolution Funding Corporation's bonds matures in April 2030 (76 FR 49477). In the window since 2022 the strip curve's 5-year point lies within the strips' maturities and its 10-year point beyond them (Internet Appendix Section B.2.2). In levels the advantage falls by 0.73 basis points for each percentage point of GDP, with an interval from −1.11 to −0.35, and on federal debt held by the public by 0.77. In 12-month changes the loading of the advantage on Treasury debt relative to GDP is −1.25, with an interval from −3.52 to 1.02 (Internet Appendix Section B.2.3). From April 2013 its loading on supply in levels is −0.24 without the dealers' positions, with an interval that just contains zero, and −0.23 beside them, with an interval from −0.58 to 0.13. Germany's 5-year dollar credit default swap priced its credit at 4.3 basis points before mid-2008, 46.6 over 2008 to 2013, 12.6 over 2014 to 2021 and 11.1 from 2022 to June 2024 (Internet Appendix Section B.2.4). The contract on the United States priced its credit at 2.3, 33.0, 12.5 and 26.4 basis points over the same eras. With both prices removed, since 2022 the Treasury's advantage over KfW is 23.9 basis points, against 8.6 as quoted on the same 30 months. As quoted on those months the excess of the Treasury yield over the benchmark yield is 22.6 to 25.7, with a set from 20.3 to 28.1. From the months of 2011 to 2013 to the months since 2022 the advantage

as quoted narrowed by 19.8 basis points, with a confidence set from −29.7 to −11.6. With each borrower's credit price removed it widened by 22.6, with a set from 10.5 to 34.8. Over those months Germany's dollar credit default swap fell by 44.4 basis points. On the same months and bootstrap draws as Internet Appendix Table IA.30, its confidence set runs from −68.9 to −22.0, and the contract on the United States changed by −2.0 basis points, with a set from −11.2 to 6.6.

Does the gap depend on the expected short end that I take on the day? In Table S.10 I replace the short-end advantage on the day with its mean over the following 60 months, in the three eras whose windows end by June 2026 (Internet Appendix Section B.2.5). The realized short end exceeds the short end on the day by 6.7 to 7.9 basis points before 2008, whose windows run through the crisis, and falls short of it by 4.5 to 7.3 over 2008 to 2013. In every era and on both agencies the gap on the realized path takes the sign of the gap on the day. One episode can dominate an era's forecast error, so the comparison is a check under rational expectations and not a test of the expected path I assume. In Table S.11 I give the 10-year accounting over 2014 to 2020 on the month-end synthetic OIS rate built from traded prices (Internet Appendix Section B.2.6). On the 76 month-ends of 2014 to 2020 with a synthetic rate, the OIS swap spread averages −30.4 basis points on the series of record and −27.3 on the synthetic rate. On the two screens of the LIBOR prints of Internet Appendix Section A.2.2, the shift is +0.24 or +0.18, with standard errors of 0.28 and 0.25.

Table S.9. The breaks in the Treasury's yield advantage

In basis points, at 5 years. The breaks in the mean of the Treasury's advantage $y^m - y$ found by the procedure of Bai and Perron (2003), with the number of breaks chosen by the Bayesian information criterion, a minimum segment of 15% of the months and at most five breaks, and the mean of each segment; a break month is the last month of its segment.

Comparator	Months	Breaks	Segment means
Home Loan Banks	219	Mar 2006, Feb 2010, Dec 2013	34.2, 60.7, 20.2, 6.2
Farm Credit	243	Apr 2006, Oct 2010, Dec 2020	37.6, 51.2, 13.3, −0.5
EIB and KfW	183	Sep 2013, Jan 2019	38.6, 17.4, 5.0

Table S.10. The gap at 5 years on the realized path of the short end

In basis points. The gap on the realized path is the Treasury's advantage at 5 years less the mean of its advantage at the short end on the same issuers over the following 60 months, on the months whose window ends by June 2026 and holds the short end in at least 48 months. Beside it are the gap on the day of Internet Appendix Table IA.31 on the same months and the mean of the realized short end less the short end on the day. The windows overlap by up to 59 months, so I give no interval; the era since 2022 has no complete window.

	Feb 2002 to Jun 2008	Jul 2008 to Dec 2013	Jan 2014 to Dec 2021	Third less first era
Home Loan Banks				
Months	48	55	75	
Gap on the realized path	12.3	32.7	5.9	−6.4
Gap on the day, same months	20.6	25.4	5.3	
Realized short end less the short end on the day	7.9	−7.3	−0.6	
Farm Credit				
Months	58	48	81	
Gap on the realized path	17.1	27.9	8.3	−8.7
Gap on the day, same months	23.7	23.4	6.5	
Realized short end less the short end on the day	6.7	−4.5	−1.5	

S.4 What Every Funded Borrower Pays Relative to a Swap

This section extends Internet Appendix Section B.3 with the detail of its checks and the samples behind two of them. The monthly funding commitment price of Internet Appendix Section B.3.2 exists in 68 of the 86 issue months and averages 6.21 basis points a year. In levels it rises with the Treasury yield less the OIS rate, by 0.12 basis points a year for each basis point, and with the difference between the two curves' survey premia. In March 2020 it was 5.95 basis points a year, close to its mean, and over 2023, on only two cells, it averaged 9.37.

The margin of the Treasury's floating-rate note at auction (Internet Appendix Section B.3.5) rises with the bill's convenience by 0.42 basis points per basis point. Restated over general collateral, the margin averages −2.7 basis points over 2014 to 2021 and 3.8 since 2022, neither distinguishable from zero at the 5% level against the fixed-b critical values. Over 2014 to 2021 the bracket from it to the margin over the bill contains the agencies' price with the index held at SOFR. On each era's own notes that price lies above the bracket's upper end by 1.7 to 1.9 basis points. Over SOFR, from April 2018, the note's margin differs from the agencies' spread at 2 years by 3.4 and −2.4 basis points on the months they share, each smaller than its standard error.

Table S.11. The OIS rate less the benchmark at 10 years on the synthetic OIS rate, 2014 to 2020

In basis points. $s - d$, the dealers' charge net of the comparator's premium, over 2014 to 2020 on the months with a month-end synthetic OIS rate built from traded prices (Internet Appendix Section A.2.2), on the series of record and on the synthetic rate, by comparator and measure of the funding commitment price as in Internet Appendix Table IA.32. Inference as in Internet Appendix Section A.9.

			$s - y$				$s - d$	
Comparator	Measure	Months	Record	Synthetic	Shift	$\hat{\chi}$	Record	Synthetic
EIB and KfW	Index at SOFR	76	−30.4	−27.3	+3.1	17.3	−20.1 to −7.9	−17.0 to −4.8
	Own notes						0.8 to 21.9	3.9 to 25.0
	Projection						−8.5 to −1.2	−5.4 to 1.9
Farm Credit	Index at SOFR	71	−30.7	−27.3	+3.4	40.1	−43.2 to −31.0	−39.8 to −27.6
	Own notes						−22.3 to −1.1	−18.9 to 2.2
	Projection						−31.5 to −24.3	−28.2 to −20.9
Home Loan Banks	Index at SOFR	35	−31.5	−31.0	+0.5	33.1	−36.9 to −24.8	−36.4 to −24.3
	Own notes						−16.1 to 5.1	−15.6 to 5.6
	Projection						−25.3 to −18.1	−24.8 to −17.6

In the regression of the within-issuer basis (Internet Appendix Section B.3.6), the interaction of the fixed-rate indicator with maturity is −0.40 basis points per year, with a standard error of 0.36. In every era the basis therefore rises by 1.2 basis points from 2 to 5 years. At 3 years the basis is −1.6 over 2019 to 2021 and 3.7 since 2022. Since 2022 the within-issuer basis lies above $s - d$ on EIB and KfW over the same months under every measure of the funding commitment price, by 5.5 to 5.7 basis points at 2 years and by 4.3 to 7.5 at 5. Part of the gap is the difference between the two curves, since the federal funds OIS rate on which I form $s - d$ has stood 1.6 basis points below the SOFR OIS rate at 2 years and 2.7 at 5 since 2022.

In the components of the signed imbalance (Internet Appendix Section B.3.7), before 2008 the agencies' book, Fannie Mae's alone until December 2008, was net paying fixed by $73.7 billion on average from January 2003, and both composites are positive. Over 2014 to 2021 it was net receiving fixed by $51.7 billion, the pension state lay on the receiving side, and both composites are negative.

The TreasuryDirect record contains 153 floating-rate note auctions from 29 January 2014 to 29 July 2026. In Internet Appendix Section B.3.5 I leave out two of $25 million, on 6 December 2019 and 14 July 2022, and the auction of July 2026, after the last era. In Table S.12 I give, by year, the issuer-month cells of the Home Loan Banks and Farm Credit that hold both a fixed-rate and a floating-rate note. These are the cells on which I measure the within-issuer basis (Internet Appendix Section B.3.6).

Table S.12. The issuer-month cells of the within-issuer basis by year

By year, the issuer-by-issue-month cells of the Home Loan Banks and Farm Credit that hold both a fixed-rate and a floating-rate note, those among them with a fixed-rate note within a quarter of a year of 2 or 3 years or within half a year of 5, those with a floating-rate note within a quarter of a year of 2 years, and the floating-rate notes of more than 3.5 years in them (Internet Appendix Table IA.39).

Year	Cells with both types	Fixed-rate near 2 years	Fixed-rate near 3 years	Fixed-rate near 5 years	Floating-rate near 2 years	Floating-rate notes above 3.5 years
2019	11	8	10	3	9	0
2020	12	12	11	5	11	0
2021	12	10	6	9	12	2
2022	12	10	10	9	12	4
2023	17	15	16	6	17	0
2024	24	18	17	18	24	4
2025	24	15	17	14	24	1
2026	16	9	8	7	16	7

S.5 The Long-Run Rate, Forecasts and Policy News

This section extends Internet Appendix Section B.4 with the long-run rate by construction, the forward gap on each construction, the remaining reports of the long-run rate and the detail of the policy announcements. Before each FOMC meeting the New York Fed's Survey of Primary Dealers asks each dealer for its estimate of the longer-run target federal funds rate. I take the median across dealers from the 115 releases of June 2012 to September 2026. In Table S.13 I give the distance of each curve's long-run rate from each report by construction (Internet Appendix Section B.4.1). In Table S.14 I give the Treasury forward less the OIS forward on each construction of the 5-year, 5-year forward (Internet Appendix Section B.4.2). In Table S.15 I set the forward gap against the long-run rates of the Summary of Economic Projections and of the Blue Chip long-range survey, beside the primary dealers' rate of Internet Appendix Table IA.46.

For the OIS forward I bootstrap the zero curve from the federal funds OIS rates at 1 to 10 years. The OIS zero curve exists in 198 months from July 2008. In 18 months of 2011 and 2012 no day within seven days of the month-end has every node the bootstrap needs. On 35 of the days the bootstrap uses, one to a month and all from July 2008 to August 2011, the nodes of 6 to 9 years are interpolated. The surveys' monthly path extends the spread of the funds rate over the bill forecast for the sixth quarter to years 6 to 10. On it that spread averages 7.8 and 6.7 basis points on the Home Loan Banks' and Farm Credit's months since 2022, against -0.4 on the issues of the long-range survey (Panel A of Table 4 of the paper). Adding it raises the forward gap and the sum by as much and leaves the forward agency gap unchanged.

From December 2022 to December 2024 the 5-year survey premium measured on the Treasury curve rose 38.7 basis points and the premium measured on the OIS curve rose 30.1 (Internet Appendix Section B.4.4). The part of the rise specific to the Treasury curve is 8.6, with a standard error of 13.2, and over the same window the Treasury yield rose 12.8 relative to the OIS rate. Over the same window the 5-year premium of Adrian et al. (2013) rose 84.2 basis points as published and 76.3 on the OIS curve, and that of Kim and Wright (2005) rose 18.7 and 10.8. Each rise has a standard error above 50. The part specific to the Treasury curve, the published rise less the rise on the OIS curve, is 7.9 with a standard error of 12.5 (Panel B of Table 4 of the paper).

Two samples on the dissemination tape hold the 20 announcements that my regressions require, both at 10 years (Internet Appendix Section B.4.9). On them the synchronized one-day response of the spread lies 7.8 and 8.7 standard errors from the one-day response at the close. On the 25 announcements the response at the close is 2.38, with a standard error of 0.67, distinguishable from zero at the 5% level. The synchronized response is −2.86, with a standard error of 6.55, and its 95% interval contains both zero and the response at the close. On the eight cells below that minimum, with 6 to 16 announcements, the synchronized standard errors are up to 36 times those at the close. On the same 25 announcements the two-day response of the spread is 1.27, with a standard error of 0.39, distinguishable from zero at the 1% level and 26% of the two-day response of the OIS rate of 4.89.

I also form the correction to the quote time from the prints within 10 basis points of the day's median. Its standard deviation on other weekdays is then 1.9 basis points at 5 years and 1.6 at 10, against 20.0 and 21.6 on the prints cleared through a central counterparty. On the 16 announcements at 10 years that the screen leaves, the synchronized response of the spread is 1.42, half the 2.84 at the close and near the 1.18 on two days. On all 69 announcements of 2013 to 2020 the two-day response of the 10-year spread is 0.58, with a standard error of 0.39, which is not distinguishable from zero at the 10% level. On the 63 scheduled announcements it is 0.49, with a standard error of 0.47, so the 25 announcements of the check do not represent their era.

Around the purchase announcements of 2008 to 2010 the 5-year LIBOR swap rate less the Treasury yield rose by 7.38 basis points on 18 March 2009 and fell by 16.04 over the five dates of the first program. It fell because the term spread of the 5-year LIBOR swap over the OIS rate fell by 21.04 over the five dates, with falls of 8.40, 8.01 and 9.22 on 25 November, 16 December and 18 March. Over the same five dates 3-month LIBOR less the 3-month OIS rate fell by 10.25 basis points (Internet Appendix Section B.4.8). In Table S.16 I give the freshness of the quotes around the eight dates. In Table S.17 I set my sums beside the changes that Krishnamurthy and Vissing-Jorgensen (2011) print for the same windows.

Table S.13. The long-run rate on the two curves: Δ by construction

In basis points. Δ of Panel A of Internet Appendix Table IA.43, the root mean squared distance of the Treasury reading from the anchor less that of the OIS reading, under the main construction, with the Treasury forward from the zero curve bootstrapped from the constant-maturity yields, with the expected funds-bill spread set to zero, without the anchor dates that use the last survey's expected spread carried forward, and, for Blue Chip, matched to the month-end after the issue's responses.

Δ, *Treasury less OIS, by construction*				
Anchor and premium	Construction	Jan 2014 to Jun 2026	Jan 2014 to Dec 2021	Jan 2022 to Jun 2026
SEP median, ACM	As main-text Table 4	10.5	−21.1	54.0
	Constant-maturity curve	7.4	−16.9	43.2
	Funds-bill spread zero	7.5	−20.0	46.7
	Without carried values	7.6	−21.1	58.1
SEP median, KW	As main-text Table 4	−7.5	−37.5	58.2
	Constant-maturity curve	−8.7	−32.3	48.5
	Funds-bill spread zero	−9.5	−35.8	51.5
	Without carried values	−15.4	−37.5	51.9
Dealer survey median, ACM	As main-text Table 4	12.4	−16.5	49.0
	Constant-maturity curve	8.3	−13.8	38.1
	Funds-bill spread zero	9.3	−15.7	42.1
	Without carried values	10.7	−16.5	54.6
Dealer survey median, KW	As main-text Table 4	−9.4	−37.8	52.3
	Constant-maturity curve	−11.4	−33.5	42.3
	Funds-bill spread zero	−11.5	−36.1	45.6
	Without carried values	−16.3	−37.8	48.2
Blue Chip, years 6 to 10, ACM	As main-text Table 4	42.7	22.9	57.9
	Constant-maturity curve	32.6	21.2	42.6
	Funds-bill spread zero	38.6	23.1	51.3
	Without carried values	42.7	22.9	57.9
	Matched to the month-end after	41.6	19.8	57.5
Blue Chip, years 6 to 10, KW	As main-text Table 4	12.7	−23.7	46.3
	Constant-maturity curve	4.2	−21.7	31.6
	Funds-bill spread zero	9.1	−23.7	40.6
	Without carried values	12.7	−23.7	46.3
	Matched to the month-end after	11.0	−23.1	47.6

Over the five dates of the first program the noncallable agency yield of Fannie Mae fell by 150 basis points at 5 years and 200 at 10, against 74 and 107 for the Treasury yield. The Aaa corporate yield fell by 88 and 77.

Table S.14. The 5-year, 5-year forward gap on each construction

In basis points. By era, the Treasury forward less the OIS forward, $2z_{10} - z_5$ on continuously compounded zero rates or $2y_{10} - y_5$ on par yields, on each construction of Internet Appendix Table IA.44, with standard errors in parentheses and the months below, and the two forwards' levels in percent on the zero curves. The fitted par yields exist from January 2013. Over January 2013 to June 2026 the gap on zero curves and the gap on fitted par yields correlate at 0.95 in levels and 0.65 in monthly changes. Inference as in Internet Appendix Section A.9.

Construction	Jul 2008 to Dec 2013	Jan 2014 to Dec 2021	Jan 2022 to Jun 2026	Jan 2013 to Jun 2026
Zero curves: off-the-run Treasury and bootstrapped OIS	66.9 (15.3)	40.9 (4.9)	61.2 (6.8)	46.5 (5.1)
	[48]	[96]	[54]	[162]
Zero curves: constant-maturity Treasury and bootstrapped OIS	29.4 (3.1)	36.8 (5.0)	50.9 (8.5)	40.3 (5.1)
	[48]	[96]	[54]	[162]
Quoted par yields (main-text Figure 2)	29.0 (3.0)	37.6 (4.8)	51.4 (7.7)	40.9 (4.9)
	[48]	[96]	[54]	[162]
Fitted par yields	22.6 (1.7)	38.3 (4.9)	51.7 (8.3)	41.6 (5.0)
	[12]	[96]	[54]	[162]
Treasury forward, zero curve (%)	4.72	2.54	4.12	3.16
OIS forward, zero curve (%)	4.05	2.13	3.51	2.69

Table S.15. The long-run rates of the FOMC projections and the Blue Chip survey and the forward gap

I regress each anchor A_t, in percent, on L_t, the long-run rate read from the OIS curve (the OIS 5-year rate 5 years ahead less the published forward premium of Adrian et al. (2013), ACM, or Kim and Wright (2005), KW), and on G_t, the forward gap on the curve of Gürkaynak et al. (2007), both in percent: $A_t = a + bL_t + cG_t + e_t$. The anchors are those of Internet Appendix Table IA.43, here the Summary of Economic Projections and the Blue Chip long-range survey, on their dates from January 2014 to June 2026, in levels and in changes between consecutive dates. The anchors are consistent with the OIS reading if $c = 0$ and with the Treasury reading if $c = b$. Because $L_t + G_t = M_t$, the Treasury forward less the published premium, which is the model's expected short rate five to ten years ahead up to its fitting error, the regression is also $A_t = a + bM_t + (c - b)G_t + e_t$: $c - b$ is the loading on the forward gap with the model's expected rate held fixed, zero for an anchor that follows the Treasury reading and $-b$ for one that follows the OIS reading, and its standard error comes from the joint covariance of b and c. Standard errors in parentheses are Newey-West at the Schwert (1989) lag for the dates, and intervals are fixed-b 95% intervals of Kiefer and Vogelsang (2005). MDD is the minimum detectable c at 80% power. The SEP median changes on 18 of its 48 consecutive dates. Internet Appendix Table IA.46 gives the dealer survey.

Anchor	Premium	Form	Dates	b	c	Interval, c	$c - b$	Interval, $c - b$	MDD
SEP median	ACM	levels	49	−0.27 (0.15)	−0.07 (0.66)	[−1.82, 1.67]	0.19 (0.54)	[−1.24, 1.63]	2.30
		changes	48	0.04 (0.05)	0.11 (0.07)	[−0.07, 0.30]	0.07 (0.06)	[−0.09, 0.24]	0.24
	KW	levels	49	0.15 (0.20)	−0.37 (0.72)	[−2.27, 1.53]	−0.51 (0.84)	[−2.73, 1.70]	2.50
		changes	48	0.10 (0.05)	0.09 (0.08)	[−0.13, 0.30]	−0.01 (0.08)	[−0.20, 0.19]	0.29

Table S.15. The long-run rates of the FOMC projections and the Blue Chip survey and the forward gap, continued

Anchor	Premium	Form	Dates	b	c	Interval, c	$c-b$	Interval, $c-b$	MDD
Blue Chip, years 6 to 10	ACM	levels	13	0.43 (0.09)	1.07 (0.32)	[−0.16, 2.30]	0.64 (0.40)	[−0.89, 2.18]	1.50
		changes	12	0.39 (0.06)	1.17 (0.36)	[−0.27, 2.61]	0.78 (0.32)	[−0.50, 2.06]	1.75
	KW	levels	13	0.45 (0.10)	0.66 (0.43)	[−0.98, 2.30]	0.21 (0.53)	[−1.80, 2.22]	2.00
		changes	12	0.25 (0.06)	0.60 (0.36)	[−0.83, 2.03]	0.36 (0.36)	[−1.07, 1.78]	1.73

Table S.16. The freshness of the quotes around the purchase announcements of 2008 to 2010

For each daily series, the trading days on which it is quoted, the days on which the quote differs from the previous quoted day out of the days compared, and the longest run of an unchanged quote, over three windows. On each of the eight dates of Internet Appendix Table IA.50, every quote changed on the announcement day and on the day after.

Series	28 July to 31 October 2008			3 November 2008 to 31 March 2009			1 April to 31 December 2009		
	Quoted	Changed	Run	Quoted	Changed	Run	Quoted	Changed	Run
OIS rate, 5 years	70 of 70	70 of 70	1	107 of 107	107 of 107	1	197 of 197	196 of 197	2
OIS rate, 10 years	68 of 70	66 of 67	2	103 of 107	103 of 103	1	193 of 197	191 of 193	2
LIBOR swap, 5 years	70 of 70	70 of 70	1	107 of 107	107 of 107	1	197 of 197	197 of 197	1
LIBOR swap, 10 years	70 of 70	70 of 70	1	107 of 107	107 of 107	1	197 of 197	197 of 197	1
Treasury yield, 5 years	68 of 70	65 of 68	2	101 of 107	99 of 101	2	189 of 197	173 of 189	2
Treasury yield, 10 years	68 of 70	66 of 68	2	101 of 107	95 of 101	2	189 of 197	180 of 189	3

Table S.17. The purchase announcements of 2008 to 2010: the sums and every trading day

In basis points. Panel A sums the two-day changes in the OIS rate, the Treasury yield and the OIS swap spread of Internet Appendix Table IA.48 and in the LIBOR swap spread of Internet Appendix Table IA.50 over the five dates of the first program, over 10 August and 21 September 2010 and on 3 November 2010, beside the changes Krishnamurthy and Vissing-Jorgensen (2011) print over the same windows for the non-callable agency yield of Fannie Mae and the Aaa corporate yield, long-term at 10 years and intermediate-term at 5, and that Aaa yield adjusted for the change in credit default swap spreads; their Treasury changes equal mine in every cell. Panel B gives the coefficient on a dummy for the eight announcements in a regression of the change in the OIS swap spread on every trading day of 2008 to 2011, with its heteroskedasticity-robust (HC1) standard error in parentheses and its wild-bootstrap p-value (Rademacher weights, 4,999 draws), and beside them the Newey and West standard error with one lag for the two-day changes, which overlap on consecutive days, and none for the one-day changes.

Panel A: The sums, beside the changes Krishnamurthy and Vissing-Jorgensen (2011) print

Announcements	Maturity	OIS rate	Treasury yield	OIS swap spread	LIBOR swap spread	Agency yield	Aaa yield	Aaa yield, CDS-adjusted
First program, five dates	5 years	−69.00	−74	5.00	−16.04	−150	−88	−82
	10 years	−87.00	−107	20.00	17.25	−200	−77	−70
10 August and 21 September 2010	5 years	−18.25	−20	1.75	−0.46	−20	−18	−20
	10 years	−24.20	−30	5.80	4.75	−29	−23	−26
3 November 2010	5 years	−11.35	−11	−0.35	−1.13	−14	−10	
	10 years	−5.50	−10	4.50	3.60	−10	5	

Panel B: A dummy for the eight announcements, every trading day of 2008 to 2011

Window	Maturity	Coefficient	Newey-West	p-value	Trading days
One day	2 years	1.18 (1.61)	(1.61)	0.526	1,002
	5 years	0.50 (3.04)	(3.04)	0.831	997
	10 years	−0.90 (2.75)	(2.75)	0.685	720
Two days	2 years	1.25 (2.33)	(2.33)	0.627	1,002
	5 years	0.91 (2.25)	(2.25)	0.818	995
	10 years	3.94 (1.22)	(1.21)	0.012	718

S.6 Data Documentation

This section extends Internet Appendix Sections A.2 and A.3. Here I list the quotes that the screens remove, the backfill of the SOFR OIS curve, the conventions of the two legs and the quote times of the dissemination tape, and I point to the record of the raw files.

S.6.1 The Quotes the Screens Remove

With the LIBOR screen of Internet Appendix Section A.2.1 I remove an OIS quote above the same-day LIBOR swap rate of the same maturity. The screen drops 3.831% at 5 years on 14 August 2003, against a LIBOR swap rate of 3.695%, and 2.842% at 10 years on 8 December 2011, against 2.101%. The timing screen, which admits a month-end only where the last quotes of its two legs fall within seven calendar days of each other, binds at 10 years

in 2011 and 2012, when the 10-year quote appears on 45% and 3% of weekdays, against at least 96% in every year since 2013. The screens change no month at 2 or 5 years and empty five at 10 years, May, September, November and December 2011 and July 2012. The 10-year series therefore misses every month from May 2011 to November 2012 except August 2011, 18 in all.

The term quotes of general collateral repo of 31 December 2018 and of the holiday of 1 January 2019 print at 1.25% to 1.35%, against 2.5% to 2.7% on 28 December and on 2 January (Internet Appendix Section A.3). I set those of 31 December to missing, so a month-end series takes the quotes of 28 December, and the H.15 release publishes neither the bill nor a Treasury yield on 1 January.

S.6.2 The SOFR OIS Curve before May 2019

Before May 2019 Bloomberg's SOFR OIS curve is a backfill of the federal funds curve. At 1, 2, 3 and 5 years it equals the federal funds OIS rate plus a constant of 3.21, 2.32, 1.11 and −0.40 basis points on all but at most 0.5% of the days of 2009 to 2017. From 2018 it first departs from the constant on 26 April 2019 at 1 and 2 years and on 6 May 2019 at 3 and 5. At 10 years, where it begins on 28 July 2008, the offset is not constant, and over 2009 to 2017 it averages −2.83 basis points with a standard deviation of 1.06. I therefore use the SOFR curve only from May 2019.

S.6.3 The Conventions of the Two Legs

The conversion of the OIS fixed rate s to the Treasury's semi-annual bond-equivalent convention, $2[(1 + s \times 365/360)^{1/2} - 1]$, raises the rate most, by 1.9 basis points, at a rate of 2.76%. I give what the conversion does to the spread in Internet Appendix Section A.3.

S.6.4 The Quote Times of the Dissemination Tape

On the dissemination tape I compare the median print in a 30-minute window with the Bloomberg close. In Table S.18 I give, by window, the median print against the Bloomberg close, from which I place the close at 5:00 pm, and the distribution of the correction to the Treasury's quote time that I use in Internet Appendix Section B.4.9. The mean differences reflect off-market prints. The correction assumes that the basis between LIBOR and the federal funds rate does not move between the windows centered at 3:30 pm and 5:00 pm. On the two announcement days on which the tape holds basis prints in both windows, both at 10 years, the basis moves by 0.16 basis points on average and by at most 0.19.

Table S.18. The Bloomberg close and the correction to the Treasury's quote time on the dissemination tape, 2013 to 2020

Panel A compares, on every weekday of 2013 to 2020 with at least three prints in the window, the median print of the spot-starting LIBOR fixed-for-floating swap at exactly 5 or 10 years on the dissemination tape of the Depository Trust and Clearing Corporation, in a 30-minute window centered at the time shown, with the Bloomberg close of the same swap: the median absolute difference, and the mean and the standard deviation of the difference, in basis points. Panel B gives the distribution of the correction $m(d)$, the median print in the window centered at 5:00 pm less the median print in the window centered at 3:30 pm, each window holding at least three prints, on the FOMC announcement days of 2013 to 2020 and on the other weekdays, in basis points. Panel C gives $m(d)$ at 10 years on each of the 25 announcements of Panel A of Internet Appendix Table IA.51, on the announcement day and on the trading day before it; the synchronized one-day change in the OIS swap spread is its change at the close less the difference between the two.

Panel A: The median print in each window against the Bloomberg close

Maturity	Window centered at	Weekdays	Median absolute difference	Mean difference	Standard deviation
5 years	3:30 pm	1,462	0.59	4.2	29.2
	4:00 pm	1,372	0.45	4.2	29.8
	4:30 pm	1,229	0.37	5.9	33.0
	5:00 pm	723	0.28	12.9	51.1
10 years	3:30 pm	1,692	0.60	4.4	27.7
	4:00 pm	1,576	0.49	5.7	32.2
	4:30 pm	1,465	0.38	8.2	38.5
	5:00 pm	1,006	0.26	13.9	51.9

Panel B: The correction $m(d)$

Maturity	Days	Days counted	Mean	Median	Standard deviation	5th percentile	95th percentile
5 years	Announcement days	31	0.0	−0.5	3.3	−4.2	5.5
	Other weekdays	549	10.0	0.1	53.9	−7.4	108.1
10 years	Announcement days	40	−0.4	−0.4	3.5	−7.5	2.0
	Other weekdays	864	9.6	0.1	54.3	−6.5	108.7

Panel C: The correction at 10 years on the 25 announcements of the check

Announcement	On the day	The day before	Announcement	On the day	The day before
1 May 2013	−0.8	23.8	26 September 2018	−1.3	0.2
19 June 2013	5.0	0.2	20 March 2019	0.2	0.3
18 December 2013	1.8	0.5	1 May 2019	−1.0	0.0
29 January 2014	0.8	0.0	31 July 2019	−0.3	−0.1
18 June 2014	−0.7	0.0	18 September 2019	−1.3	−1.3
17 September 2014	0.5	0.2	29 January 2020	−0.8	199.3
18 March 2015	−7.3	−0.2	3 March 2020	−1.1	43.6
29 July 2015	14.1	−0.4	19 March 2020	1.8	−0.8
1 February 2017	−0.3	1.6	31 March 2020	−1.0	3.8
20 September 2017	−0.7	1.5	29 April 2020	0.1	−45.5
1 November 2017	0.6	−1.6	10 June 2020	−1.1	−0.3
31 January 2018	−0.5	−16.5	27 August 2020	1.4	0.0
21 March 2018	−1.4	10.0			

S.6.5 The Raw Files

The replication package lists every raw file that the code reads, with its size and its SHA-256 checksum, in `data/RAW_MANIFEST.txt`, and the code that reads each file names the fields it uses. The vendor files themselves are not in the package, since I am not licensed to redistribute them. A file whose checksum differs from the manifest is not the extract that produced the paper's estimates.

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